

Cybersecurity Spend Record Matching

1. Summary

The product repository and two spend files (easy and hard) were used to create a single, matched output. Text in the spend item and description fields was cleaned to a comparable form, and products were matched in two passes: an exact/substring check followed by a fuzzy fallback. Confidence was reported as 100 for exact hits and the fuzzy score for non-exact hits; rows without a reliable signal were left as no_match with confidence 0. The final deliverable was written as matched_spend_records.csv, with all original columns preserved and new matching fields added for transparency.

2. Data overview

Three CSV files were in scope:

- **Product repository** - canonical product names with vendor names where available. ●

Spend (easy) and **Spend (hard)** procurement lines that reference products in free text.

- A dataset_source flag was added so each spend row could be traced to “easy” or “hard”.

Observed shapes in the run were approximately **repo: 119×3** and **combined spend: 300×8** (values may vary slightly in other runs). These checks served as quick sanity validation before matching.

3. Approach and Reasoning

A simple, explainable pipeline was preferred so results could be reviewed easily.

- **Normalization.** Text was standardized so names became comparable: lowercasing, removal of punctuation and extra spaces, basic symbol fixes (e.g., \$→s, 0→o, @→a), and removal of obvious filler words (e.g., *license*, *support*, *renewal*, *services*).

- **Two-pass matching.**

1. **Exact/substring pass** against the normalized repository name provided the strongest signal and was scored as **100**.

2. **Fuzzy fallback** using partial_ratio captured mild typos and spacing differences;

matches at or above a threshold of **85** were accepted, and the fuzzy score was reported as confidence.

- **Transparency.** For each row, the method used (exact, fuzzy, or no_match) and the confidence were recorded. Multiple products per row were retained as lists to avoid discarding potential signals.

4. Challenges identified in the dataset

Several characteristics of the spend text increased matching difficulty:

- **Typos and near-spellings** (e.g., *crowdstrik*, *zcaler*).
- **Abbreviations and short forms** (e.g., *Zscaler IA*, *CS falcon*).
- **Procurement filler words** surrounding product names (e.g., *renewal*, *subscription*, *annual*).
- **Multiple products** mentioned in a single line, requiring list outputs.
- **Vague descriptions** without a distinct product reference, which were intentionally left as no_match.

5. Handling ambiguous or partial matches

Ambiguity was handled through cautious rules and clear labeling:

- **Exact precedence.** A normalized substring of the product name in the spend text was treated as a definitive hit (confidence **100**).
- **Fuzzy acceptance with guardrails.** When no exact hit existed, partial_ratio was used with a threshold of **85** to accept only reasonably close matches.
- **Multiple candidates.** When several products were plausible, all candidates were retained in list columns; the best score among them was surfaced as the row-level confidence.
- **No forced matches.** Where the text was too weak or too generic, the row remained no_match with confidence **0** to keep outcomes honest and auditable.
- **Borderline review.** Fuzzy cases in the 80–92 band were considered borderline and suitable for manual review in a production process.

6. Results overview

Key outcomes from the matched file are summarized below. Values can be copied directly from the notebook's QA cell.

- **Total spend rows processed:** [TOTAL_ROWS]
- **Unmatched rows:** [UNMATCHED_ROWS]
- **Method distribution:** exact = [N_EXACT], fuzzy = [N_FUZZY], no_match = [N_NOMATCH]
- **Top matched products (by count):** [list 5–10 products with counts]

A small preview of the final columns can be seen in the first few rows of the output, which show the original fields plus product_id, product_name, vendor_name, confidence, and method.

7. Ambiguity examples (contextual illustrations)

A few representative rows help explain the scoring and labels:

- **Clear exact:** Text contained “Zscaler Internet Access”; confidence **100**; method **exact**.
- **Good fuzzy:** Text contained “Crowdstrik falcon”; confidence [e.g., **94**]; method **fuzzy**.
- **No match:** Text referenced “Security tools – annual renewal” with no product name; confidence **0**; method **no_match**.

8. QA checks performed

Light checks were included to keep the process reliable:

- **File path verification** (OK/MISSING) before reading.
- **Shape prints** after loading to confirm expected row counts.
- **Source tagging** (dataset_source) for traceability.
- **Spot checks** on fuzzy cases and the first few rows of the final output.

9. Deliverables & how to reproduce

Deliverables

- matched_spend_records.csv - required output with all fields.
- matched_spend_records_pretty.csv - optional Excel-friendly copy (lists collapsed with |).
- Notebook (.ipynb) - step by step code and comments.
- This report (PDF).

Reproduction steps

1. Open the notebook (Colab or Jupyter).
2. Mount Google Drive (Colab).
3. Set file paths to the three input CSVs.
4. Run cells top-to-bottom: normalize → match → export.
5. Download or retrieve the saved CSV from Drive.

10. Assumptions & limitations

- Repository product_name is treated as the canonical label.
- Vendor presence in the repository is not guaranteed and is not required for the baseline.
- No vendor-specific alias or bundle rules are hard-coded in this version.
- The fuzzy threshold of **85** was chosen as a practical balance and may be tuned for different datasets.

Appendix A - Normalization details

- **Stop-words removed:** license, licenses, licence, licences, renewal, support, subscription, subs, annual, year, via, and, for, the, of, qty, ea, each, service, services.
- **Symbol fixes:** \$→s, 0→o, @→a, 1→i, 5→s, 3→e.
- **Whitespace/punctuation:** punctuation stripped and multiple spaces collapsed.

Appendix B - Parameter choices • Fuzzy

metric: rapidfuzz.partial_ratio. • **Threshold:** 85.

- **Exact match confidence:** 100.

- **No match confidence:** 0.